

Unit 6 Pre-Test

Functions & Linear Relationships

Standards assessed: NC.8.F.1–F.5 · NC.8.EE.8

Name: _____ Date: _____ Period: _____

SCORE **3.0** Basic Understanding

NC.8.F.1 Identify functions and read slope and y-intercept from an equation.

• I can decide whether a relation is a function and read slope and y-intercept from $y = mx + b$.

Match each vocabulary term to its definition, then answer each question.

B ▾ function ✓	A. Where a graph crosses the y-axis (the value of y when $x = 0$).
D ▾ slope ✓	B. A relation in which each input has exactly one output.
A ▾ y-intercept ✓	C. A point written (x, y) that gives a location on the graph.
C ▾ ordered pair ✓	D. The steepness of a line, found as rise over run.
E ▾ linear function ✓	E. A function whose graph is a straight line.
F ▾ input ✓	F. The x-value that is put into a function.

Slope-intercept form — reference

$$y = mx + b$$

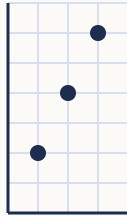
m = slope (rise over run) · b = y-intercept (where $x = 0$)

1. Function?

Is this a function? (1, 2), (2, 4), (3, 6)

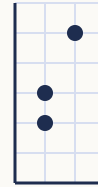
2. Function?

Is this a function? (1, 2), (1, 3), (2, 5)



Answer:

answer: yes

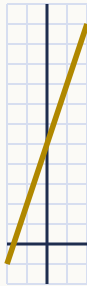


Answer:

answer: no

3. Slope

In $y = 3x + 5$, what is the slope?



$m =$

answer: 3

4. y-intercept

In $y = 3x + 5$, what is the y-intercept?

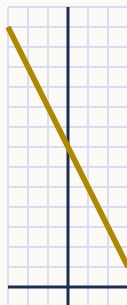


$b =$

answer: 5

5. Slope

In $y = -2x + 7$, what is the slope?

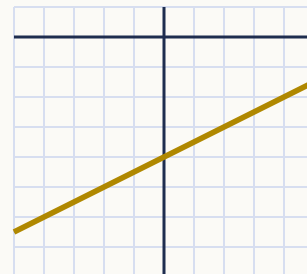


$m =$

answer: -2

6. y-intercept

In $y = \frac{1}{2}x - 4$, what is the y-intercept?



$b =$

answer: -4

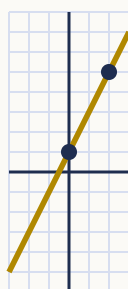
NC.8.F.4 Find slope and y-intercept from two points and from a table.

• *I can find the slope and y-intercept from two points or a table of values.*

Find each slope or y-intercept.

7. Slope

Find the slope through (0, 1) and (2, 5).

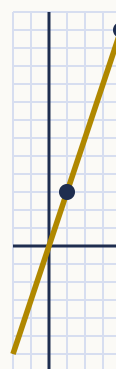


$m =$

answer: 2

8. Slope

Find the slope through (1, 3) and (4, 12).

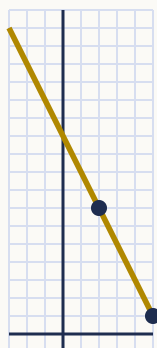


$m =$

answer: 3

9. Slope

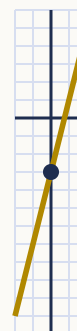
Find the slope through (2, 7) and (5, 1).



$m =$

10. y-intercept

A line has slope 4 and passes through (0, -3). Find the y-intercept.



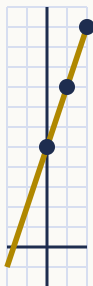
$b =$

answer: -3

answer: -2

11. From a table

x : 0, 1, 2 and y : 5, 8, 11. Find the slope.

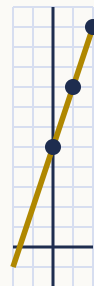


$m =$

answer: 3

12. From a table

x : 0, 1, 2 and y : 5, 8, 11. Find the y -intercept.



$b =$

answer: 5

SCORE 5.0 Advanced Content

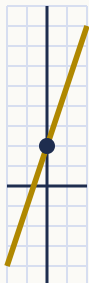
NC.8.F.2 • NC.8.EE.8 Write equations, compare functions, interpret slope, and solve a system by graphing.

• I can write a linear equation, compare two functions, interpret slope as a rate of change, and read the solution of a system from its graph.

Answer each question.

13. Write equation

A line has slope 3 and passes through (0, 2). Write $y = mx + b$.



14. Write equation

A line has slope -4 and passes through (0, -1). Write $y = mx + b$.



$$y = \underline{\hspace{2cm}}$$

answer: $y = 3x + 2$

15. Compare

Function A: $y = 2x + 3$. Function B: slope 5. Which has the greater slope? (A or B)



Answer: $\underline{\hspace{2cm}}$

answer: B

17. Slope

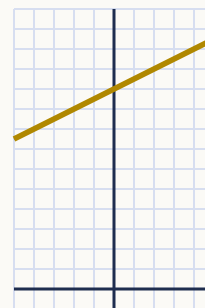
Find the slope through (1, 5) and (3, 11).

$$y = \underline{\hspace{2cm}}$$

answer: $y = -4x - 1$

16. Rate

In $y = 0.5x + 10$, how much does y increase when x goes up by 1?

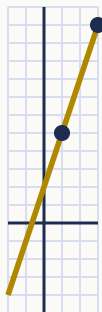


= $\underline{\hspace{2cm}}$

answer: 0.5

18. y-intercept

A line has slope 2 and passes through (3, 11). Find the y-intercept.

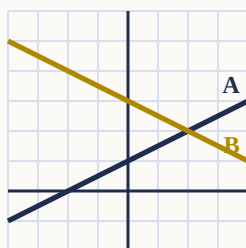


m =

answer: 3

19. System

Lines A and B are graphed. Write the solution of the system as an ordered pair.



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answer: (2, 2)

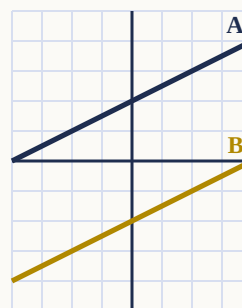


b =

answer: 5

20. System

Lines A and B are graphed. How many solutions does this system have?



Answer:

answer: none